Elementary Analysis

Section 38: Continuous Nowhere-Differentiable Functions

Exercises 38.1–38.6, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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38 Continuous nowhere-differentiable functions

No point of a nowhere differentiable function has a finite derivative. Such a function can nevertheless be continuous. Section 38 constructs examples with both properties. When an exercise needs such a function, Theorem 38.1 supplies a bounded continuous example on $\mathbb{R}$.

One useful strategy is to suppose a derivative exists and use derivative rules to force a derivative of the known nowhere-differentiable function. This is a contradiction argument: the rules are applied only to functions that are differentiable under the temporary assumption, never directly to a function already known to be nondifferentiable.

In the metric-space notation used in this section, $C(\mathbb{R})$ consists of *bounded* continuous functions, and $d(f,g) = \sup_{x\in\mathbb{R}} |f(x) - g(x)|$. The notation C^∞ instead means that derivatives of every finite order exist and are continuous. These are two different uses of the letter C.

38.1 Exercise 38.1: Why a Taylor series cannot produce the desired example

What we must show. Explain the obstruction to constructing a continuous nowhere-differentiable function as a convergent Taylor series.

Recall what a positive-radius power series guarantees.

Suppose

$$f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n$$

has radius of convergence $R > 0$. On the open interval $(c-R, c+R)$, the power-series differentiation theorem gives

$$f'(x) = \sum_{n=1}^{\infty} n a_n (x-c)^{n-1}.$$

In particular f is differentiable at every interior point of its convergence interval, however complicated the coefficients might be. Repeating the theorem gives derivatives of every order there.

Check the possible loopholes.

Behavior at the two endpoints of a finite convergence interval can be different, but those endpoints do not remove differentiability at all the interior points. A radius of convergence $R = 0$ gives only a value at the center, not a function on an interval. Moreover, a literal Taylor series of a function requires derivatives at its center in order to define the coefficients in the first place.

Conclusion. A positive-radius Taylor series is necessarily differentiable on its interior, so it cannot be a nowhere-differentiable function.

38.2 Exercise 38.2: A polynomial derivative forces a polynomial function

What we must show. If $f^{(n)}$ is a polynomial on $\mathbb{R}$ for some nonnegative integer n, prove that f is a polynomial.

Build a polynomial with the prescribed nth derivative.

The case $n = 0$ already says that f is a polynomial. For $n \geq 1$, write

$$f^{(n)}(x) = p(x) = \sum_{k=0}^{m} a_k x^k, \qquad P(x) = \sum_{k=0}^{m} a_k \frac{k!}{(k+n)!} x^{k+n}.$$

Differentiating x^{k+n} exactly n times multiplies it by $(k+n)!/k!$ and leaves x^k. Hence $P^{(n)} = p$.

Subtract and use the zero-derivative principle.

Let $h = f - P$. Then $h^{(n)} = 0$ everywhere. Proposition 38.4 implies that h is a polynomial of degree at most $n-1$. To recall why: $h^{(n-1)}$ has derivative zero, so the Mean Value Theorem makes it constant. Subtract an antiderivative of that constant from $h^{(n-2)}$; the resulting derivative is zero, so that difference is constant as well. Repeating gives a polynomial expression for h after n steps.

Conclusion. $f = P + h$ is a sum of polynomials and is therefore a polynomial.

38.3 Exercise 38.3: A differentiable function with a nowhere-differentiable derivative

What we must show. Construct a differentiable function whose first derivative is nowhere differentiable.

Integrate a continuous nowhere-differentiable function.

Let w be a continuous nowhere-differentiable function on $\mathbb{R}$, whose existence is proved in this section. Define

$$H(x) = \int_0^x w(t)\, dt.$$

For every real x, this is an ordinary integral over a finite interval (with reversed orientation if $x < 0$), so it exists.

Use the hypothesis that makes the Fundamental Theorem applicable.

Since w is continuous at every x, the Fundamental Theorem of Calculus gives $H'(x) = w(x)$ at every real x. Thus H is differentiable everywhere, and H' is nowhere differentiable. Equivalently, H has no second derivative at any point.

Conclusion. An antiderivative of a continuous nowhere-differentiable function is the required example.

38.4 Exercise 38.4: Adding a differentiable function cannot repair a derivative

What we must show. If f is nowhere differentiable and g is differentiable on $\mathbb{R}$, decide whether $f + g$ must be nowhere differentiable.

Suppose there were an exceptional point.

Assume, for contradiction, that $f + g$ is differentiable at some a. By hypothesis g is differentiable at that same a. The difference of two functions differentiable at a point is differentiable there. Thus

$$f = (f + g) - g$$

would be differentiable at a, with derivative $(f+g)'(a)-g'(a)$. This contradicts nowhere differentiability of f.

Conclusion. Yes. The sum $f + g$ is nowhere differentiable.

38.5 Exercise 38.5: Two rough functions can cancel in a sum or a product

(a) A sum need not remain nowhere differentiable.

Take any continuous nowhere-differentiable function w and let $f = w$, $g = -w$. The function g is also nowhere differentiable: if $g'(a)$ existed, multiplying by -1 would give $w'(a)$. Both are continuous, but $f + g = 0$, which is differentiable everywhere.

(b) Arrange a nonzero function before taking its reciprocal.

Choose a bounded continuous nowhere-differentiable w and a finite bound $M \geq 0$ with $|w(x)| \leq M$. Define

$$f(x) = 1 + \frac{w(x)}{2(M+1)}.$$

This is continuous, satisfies

$$d(f, 1^*) \leq \frac{M}{2(M+1)} < \frac{1}{2} < 1, \qquad \frac{1}{2} < f(x) < \frac{3}{2},$$

and is nowhere differentiable: differentiability of f would imply differentiability of $w = 2(M+1)(f-1)$ at the same point.

Verify that the reciprocal is itself nowhere differentiable.

Let $g(x) = 1/f(x)$. It is continuous because f never vanishes. Suppose g were differentiable at a point a. We have $g(a) = 1/f(a) \neq 0$. The reciprocal differentiation rule, applied to this temporarily differentiable function g, would then make $1/g = f$ differentiable at a, with derivative $-g'(a)/g(a)^2$. This contradicts the established property of f. Thus g is nowhere differentiable too, yet $f(x)g(x) = 1$ everywhere.

Conclusion. Neither a sum nor a product of two continuous nowhere-differentiable functions must be nowhere differentiable.

38.6 Exercise 38.6: A quotient of smooth functions is smooth

What we must show. Suppose $f, g \in C^\infty(\mathbb{R})$ and $g(x) \neq 0$ for every x. Prove $f/g \in C^\infty(\mathbb{R})$.

Plan. Inductively describe each derivative so that its denominator is a power of the nonvanishing function g.

State a formula suited to induction.

We claim that, for every integer $n \geq 0$, there is $h_n \in C^\infty(\mathbb{R})$ for which

$$\left(\frac{f}{g}\right)^{(n)} = \frac{h_n}{g^{2^n}}.$$

For $n = 0$, choose $h_0 = f$; the denominator is $g^{2^0} = g$, so the formula is correct.

Differentiate the inductive expression.

Suppose the formula holds for some n, and write $m = 2^n$. The function g^m is smooth, by the product rule applied a finite number of times, and it never vanishes. Thus the ordinary quotient rule, requiring only first derivatives at this step, gives

$$\left(\frac{h_n}{g^m}\right)' = \frac{h_n' g^m - h_n\, m g^{m-1} g'}{g^{2m}}.$$

Define

$$h_{n+1} = h_n' g^m - m h_n g^{m-1} g'.$$

All factors are smooth, so finite sums and products show that h_{n+1} is smooth. Since $2m = 2^{n+1}$, this is the desired formula at the next order. The induction also proves the existence of that next derivative.

Check continuity, not only existence.

For each n, the formula expresses the nth derivative as a quotient of continuous functions with a denominator that is nowhere zero. That derivative is therefore continuous. We have proved the existence and continuity of derivatives of every order.

Conclusion. f/g is infinitely differentiable. The nonvanishing assumption is what allows every quotient-rule step.